

STOCHASTIC METHODS IN WATER RESOURCES

Unit 2: Hydrological statistics and extremes

Lecture 7: Probability distributions of extremes, distribution selection

Luis Alejandro Morales, Ph.D.

Universidad Nacional de Colombia
Department of Civil and Agriculture Engineering

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Gumbel distribution

The **Gumbel distribution** is equivalent to the asymptotic distribution type I and result from a basic **exponential distribution**. The **Gumbel distribution** has two parameters: the **scale parameter** α and the **position parameter** b , and is given for **maximum values** represented by X , where:

$$f_X(x) = \frac{1}{\alpha} e^{\left[-\frac{x-b}{\alpha} - e^{\left(-\frac{x-b}{\alpha}\right)}\right]}, \text{ for } -\infty < x < \infty$$

The **cdf** is:

$$F_X(x) = e^{\left[-e^{\left(-\frac{x-b}{\alpha}\right)}\right]}$$

and $\mu_X = b + 0.5772\alpha$ and $\sigma_X^2 = \frac{\pi^2\alpha^2}{6}$ and the asymmetry coefficient $\gamma_1 = 1.1396$. The **quantile** of this distribution is:

$$x(F) = b - \alpha \ln(-\ln F)$$

where $x(F)$ is the value of X for which the **cdf** value is F . This equation is equivalent to:

$$x_T = b - \alpha \ln \left[-\ln \left(1 - \frac{1}{T} \right) \right]$$

where T is the **return period** and x_T is the magnitude of the extreme event associated to this.

Gumbel distribution

The parameter estimation in the **Gumbel distribution** can be performed using different methods such as the Moments and the Maximum-Likelihood. The **frequency factor** K_T for this distribution when $n \rightarrow \infty$, converge asymptotically to:

$$K_T = -\frac{\sqrt{6}}{\pi} \left[0.5772 + \ln \left(\ln \left(1 - \frac{1}{T} \right) \right) \right]$$

On the other hand, for a value of the annual series, the **frequency factor** as a function of the size series, can be estimated as:

$$K_m = \frac{y_m - \bar{y}}{s_y}$$

where:

$$y_m = -\ln \left[-\ln \left(\frac{n+1-m}{n+1} \right) \right]$$

Note that the data are sorted in descending order; from the largest ($m = 1$) to the smallest ($m = n$). $\bar{y} = \frac{1}{n} \sum_{m=1}^n y_m$ and $s_y^2 = \frac{1}{n} \sum_{m=1}^n (y_m - \bar{y})^2$. For a **return period** T , the **frequency factor** is:

$$K_T = \frac{y_T - \bar{y}}{s_y}$$

where $y_T = -\ln \left[-\ln \left(1 - \frac{1}{T} \right) \right]$. Comparing the two equations to compute K_T the later equations provide larger results of K_T than the former one. This value is progressively larger as much as n decrease. This means that the shorter the series the larger the values of K_T .

Gumbel distribution

Estimation of the **standard deviation** s_T using different methods:

► **Method of Moments**

$$s_T^2 = \frac{s_X^2}{n} \left[1 + K_T \gamma_1 + \frac{K_T^2}{4} (\gamma_2 - 1) \right]$$

where $\gamma_1 = 1.1396$ and $\gamma_2 = 5.4002$

► **Method of Maximum-Likelihood**

$$s_T^2 = \frac{\hat{\alpha}^2}{n} \left[1 + \frac{6}{\pi^2} (1 - 0.5772 + y_T)^2 \right]$$

► **Method of MPP**

$$s_T^2 = \frac{\hat{\alpha}^2}{n} [1.1128 + 0.4574 y_T + 0.8046 y_T^2]$$

Gumbel distribution

So far, we have analysed annual series of **maximum values** using the **Gumbel distribution**. However, this distribution can be used to analyze **minimum values** Z so the **pdf** is:

$$f_Z(z) = \frac{1}{\alpha} e^{\left[\frac{z-b}{\alpha} - e^{\left(\frac{z-b}{\alpha} \right)} \right]}, \text{ for } -\infty < z < \infty$$

and the **cdf** is:

$$F_Z(z) = 1 - e^{\left[-e^{\left(\frac{z-b}{\alpha} \right)} \right]}$$

The $\mu_Z = b - 0.5772\alpha$, $\sigma_Z^2 = \frac{\pi^2 \alpha^2}{6}$ and the asymmetry coefficient is $\gamma_1 = -1.1396$.
The **quantile** is:

$$z(F) = b + \alpha \ln(-\ln(1 - F))$$

where $z(F)$ is the value of Z for which the **cdf** value is F . This equation is equivalent to:

$$z_T = b + \alpha \ln \left[-\ln \left(1 - \frac{1}{T} \right) \right]$$

Similarly, the parameter of the distribution can be estimated using different methods.

Gumbel distribution

When $n \rightarrow \infty$ asymptotically, the frequency factor K_T converge to :

$$K_T = \frac{\sqrt{6}}{\pi} \left[0.5772 + \ln \left(-\ln \left(1 - \frac{1}{T} \right) \right) \right]$$

On the other hand, for a value of the annual series, the frequency factor as a function of the size series, can be estimated as:

$$K_m = \frac{\bar{y} - y_m}{s_y}$$

where:

$$y_m = \ln \left[-\ln \left(\frac{m}{n+1} \right) \right]$$

Note that the data are sorted in ascending order; from the smallest ($m = 1$) to the largest ($m = n$). $\bar{y} = \frac{1}{n} \sum_{m=1}^n y_m$ and $s_y^2 = \frac{1}{n} \sum_{m=1}^n (y_m - \bar{y})^2$.

For a return period T , the frequency factor is:

$$K_T = \frac{\bar{y} - y_T}{s_y}$$

where $y_T = -\ln \left[-\ln \left(-\frac{1}{T} \right) \right]$.

Pearson type III and Log-Pearson type III distributions

The pdf of the **Pearson type III distribution** is:

$$f_X(x) = \frac{1}{a\Gamma(b)} \left(\frac{x-c}{a} \right)^{b-1} e^{-\left(\frac{x-c}{a}\right)}$$

where the parameters a , b and c are the scale, the shape and the position, respectively. Note that $b > 0$ and that $c < x < \infty$. If $c = 0$, the pdf is reduced to the **Gamma distribution**. Despite a can be either positive or negative, if $a < 0$ the pdf is bounded above, this means that $x < c$ and it is not suitable to analyse maximum extreme events. Using the **methods of Moments**:

► **Mean**

$$\mu_X = c + ba$$

► **Variance**

$$\sigma_X^2 = ba^2$$

► **Asymmetry coefficient**

$$\gamma_1 = \operatorname{sgn}(a) \frac{2}{\sqrt{b}}$$

where sgn is the sign function.

Sometimes, the annual series are transformed using the logarithm function and the data is fitted to the **Pearson type III distribution**, which is equivalent to the **Log-Pearson type III distribution**. This distribution is commonly used to analyse maximum streamflows and requires that $b > 1$ and $a > 0$. The equations to estimate the frequency factor are given in textbook tables.

Weibull distribution

The **Weibull distribution** is equivalent to the asymptotic type III distribution, and is popular to the analysis of annual series of minimum values Z . The **pdf** is defined as:

$$f_Z(z) = \left(\frac{\kappa}{\alpha}\right) \left(\frac{z-b}{\alpha}\right)^{\kappa-1} e\left[-\left(\frac{z-b}{\alpha}\right)^{\kappa}\right]$$

where $z \geq b$, $\alpha > 0$ and $\kappa > 0$. The **cdf** is:

$$F_Z(z) = 1 - e\left[-\left(\frac{z-b}{\alpha}\right)^{\kappa}\right]$$

Some expressions are:

► **Mean**

$$\mu_Z = b + \alpha \Gamma\left(1 + \frac{1}{\kappa}\right)$$

► **Variance**

$$\sigma_Z^2 = \alpha^2 \left[\Gamma\left(1 + \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right) \right]$$

► **Asymmetry coefficient**

$$\gamma_1 = \frac{\Gamma\left(1 + \frac{3}{\kappa}\right) - 3\Gamma\left(1 + \frac{1}{\kappa}\right)\Gamma\left(1 + \frac{2}{\kappa}\right) + 2\Gamma^3\left(1 + \frac{1}{\kappa}\right)}{\left[\Gamma\left(1 + \frac{2}{\kappa}\right) - \Gamma^2\left(1 + \frac{1}{\kappa}\right)\right]^{1.5}}$$

where $\Gamma(\cdot)$ is the **Gamma function**.

The inverse of the **Weibull cdf** is:

$$z(F) = b + \alpha [-\ln(1 - F)]^{1/\kappa}$$

Note that when $\kappa = 1$ this distribution becomes the **exponential distributio**. Other properties of the **Weibull distribution** are given in textbooks tables.

Fréchet distribution

The **Fréchet distribution** is equivalent to the asymptotic type II distribution. The pdf is defined as:

$$f_X(x) = \frac{\theta}{a} \left(\frac{a}{x}\right)^{\theta+1} e\left[-\left(\frac{a}{x}\right)^{\theta}\right]$$

where $x > 0$, and the scale and shape parameters are $a > 0$ and $\theta > 0$, respectively.

The cdf is:

$$F_X(x) = e\left[-\left(\frac{a}{x}\right)^{\theta}\right]$$

Some expressions are:

► Mean

$$\mu_X = a\Gamma\left(1 - \frac{1}{\theta}\right)$$

valid when $\theta > 1$

► Variance & variation coefficient

$$\sigma_X^2 = a^2 \left[\Gamma\left(1 - \frac{2}{\theta}\right) - \Gamma^2\left(1 - \frac{1}{\theta}\right) \right]$$

$$V_X = \left[\frac{\Gamma(1 - 2/\theta)}{\Gamma^2(1 - 1/\theta)} - 1 \right]^{0.5}$$

where the functions for σ_X^2 and V_X are valid for $\theta > 2$.

The quantile for the **Fréchet pdf** is given:

$$x(F) = a(-\ln F)^{-\theta}$$

Regarding to the exponential positive shape for $\theta > 0$, $x(F)$ increases faster than the **Gumbel distribution** when F increase. This means that **Fréchet distribution** produces a frequency curve of larger magnitudes. These two distributions are related through the **logarithm transform**, because when X follow a **Fréchet pdf** with parameters a and θ , $Y = \ln(X)$ follow a **Gumbel pdf** with parameters $\alpha = 1/\theta$ and $b = \ln(a)$. Other properties of the **Fréchet distribution** are given in textbooks tables.

GEV distribution

The **Generalized Extreme Value (GEV) distribution** is used for extreme maximum values, usually meteorological data. The pdf is:

$$f_X(x) = \frac{1}{a} \left[1 - \frac{b}{a}(x - c) \right]^{(1-b)/b} e^{-[1 - \frac{b}{a}(x - c)]^{1/b}}$$

and the cdf is:

$$F_X(x) = e^{-[1 - \frac{b}{a}(x - c)]^{1/b}}$$

where $a > 0$, b and c are the parameters of scale, shape and position, respectively. If $b > 0$ the distribution is bounded above representing a type III distribution which is useful for the analysis of extreme minimum events. In contrast, if $b < 0$ the distribution is bounded below equivalent to a type II distribution suitable for extreme maximum events. If $b = 0$, the **GEV distribution** is transformed to a **Gumbel distribution**.

The first two moments are:

► **Mean**

$$\mu_X = c + \frac{a}{b} [1 - \Gamma(1 + b)]$$

► **Variance**

$$\sigma_X^2 = \frac{a^2}{b^2} [\Gamma(1 + 2b) - \Gamma^2(1 + b)]$$

where $\Gamma(r)$ is the **Gamma function** evaluated in r . Note that while the equation for μ_X is valid for $b > -1$, the equation for σ_X^2 is valid for $b > -0.5$. A quantile is computed as:

$$x(F) = c + \frac{a}{b} \left[1 - (-\ln F)^b \right]$$

Other properties of the **GEV distribution** are given in textbooks tables.

Pareto distribution

The **Pareto distribution** is commonly used to analysed variables with long, straight tails. This distribution was initially implemented to analysed the wealth distribution in society where a low percentage concentrate the largest wealth. The pdf classic Pareto distribution or Pareto I distribution is:

$$f_x(x) = a_l c_l^{a_l} x^{a_l-1}$$

where $x > c_l$ and $a_l > 0$. c_l and a_l are the position and shape parameters, respectively. The cdf is:

$$F_X(x) = 1 - \left(\frac{c_l}{x}\right)^{a_l}$$

The fist two moments are:

► Mean

$$\mu_X = \frac{c_l a_l}{a_l - 1}$$

Note that this is valid when $a_l > 1$, and ∞ on the contrary.

► Variance

$$\sigma_X^2 = \left(\frac{c_l}{a_l - 1}\right)^2 \frac{a_l}{a_l - 2}$$

when $a_l > 2$. If $1 < a_l \leq 2$, $\sigma_X^2 = \infty$, and not exist when $a_l \leq 1$.

Pareto distribution

The **Pareto I distribution** is embedded in a more general **Pareto II distribution**. The **pdf** of this distribution is:

$$f_X(x) = \begin{cases} \frac{1}{b} \left(1 - a \frac{x}{b}\right)^{\frac{1}{a}-1} & \text{for } a \neq 0 \\ \frac{1}{b} e^{\left(-\frac{x}{b}\right)} & \text{for } a = 0 \end{cases}$$

The **cdf** is:

$$F_X(x) = \begin{cases} 1 - \left(1 - a \frac{x}{b}\right)^{\frac{1}{a}} & \text{for } a \neq 0 \\ 1 - e^{\left(-\frac{x}{b}\right)} & \text{for } a = 0 \end{cases}$$

where b is the scale parameter. Including a third parameter c , the position parameter, the **pdf** of the **Pareto III distribution** is:

$$f_X(x) = \begin{cases} \frac{1}{b} \left(1 - a \frac{x-c}{b}\right)^{\frac{1}{a}-1} & \text{for } a \neq 0 \\ \frac{1}{b} e^{\left(-\frac{x-c}{b}\right)} & \text{for } a = 0 \end{cases}$$

The **cdf** is:

$$F_X(x) = \begin{cases} 1 - \left(1 - a \frac{x-c}{b}\right)^{\frac{1}{a}} & \text{for } a \neq 0 \\ 1 - e^{\left(-\frac{x-c}{b}\right)} & \text{for } a = 0 \end{cases}$$

Note that when $c = 0$, the **Pareto III distribution** becomes the **Pareto II distribution**. Also, when $a < 0$ and $c = 0$ becomes the **Pareto I distribution** with $a_I = 1/a$ and $c_I = -b/a$.

Frequency curves for minimum events

- ▶ The commonly implemented probability distributions to analysed annual series of minimum events are: Weibull, Gumbel, Lognormal, Pearson type III and GEV. In the case of the Gumbel distribution, which is a unbounded distribution, can cause problem because negative values of the analysed variable can arise that are not realistic as is the case of some hydrological variables.
- ▶ These distribution can be fitted to the annual series of minimum values $z_i, i = 1, \dots, n$ using a data transformation, that consist in multiplying the data by -1 , which means that $x_i = -z_i$. This means that low values becomes high values and vice-versa. Thus, the fit of the probability distribution is made based on a annual series of maximum values x_i . At the end, the quantiles estimated for the frequency curve must be multiplied by -1 to return to the domain of Z .

Fitting process

The following steps summarize the process of fitting a probability distribution to an annual series.

1. Establishment of the annual series after applying statistical tests to detect **outliers**, and verify **independence**, **homogeneity** and **stationarity** in the series. Recall, that the size of the series affect in the estimation of the **frequency curve** and in the associated uncertainty. The literature thus recommends that the number of values of the series must be 25-30.
2. Selection of the **pdf** $f_X(x)$
3. Selection of the methods for parameter estimation $\hat{\theta}$
4. Estimation of magnitudes of events for selected **return periods** T , which depends upon the **cdf** $F_X(x)$ and the estimated parameters $\hat{\theta}$. Note that the T values should be less than 3-4 times the length (number of years) of the series. For maximum events:

$$\hat{X}_T = F_X^{-1} \left[1 - \frac{1}{T} \right]$$

and for minimum events:

$$\hat{X}_T = F_X^{-1} \left[\frac{1}{T} \right]$$

5. Estimation of the **confidence limits** for X_T for a significance level α . This means, confidence limits for the frequency curve.
6. Check through Q-Q plots and plots of the frequency curve if the chosen distribution is compatible with the annual series. Also, the use of the moments can verify the correctness of the chosen distribution with respect to the annual series. For instance, when γ_1 is close to zero, the **Normal distribution** could be considered. On the contrary, other distribution could be more appropriate. Also, hypothesis tests can be applied such as the **Chi-square**, **Kolmogorov-Smirnov** and **Anderson-Darling**, to verify the goodness of fit.

Anderson-Darling test

The **Anderson-Darling test** to verify the goodness of fit of a distribution is more adequate for the analysis of extreme events, because the test give more weight to the distribution tails. To perform this test on the following distributions: **Gumbel**, **Fréchet**, **Normal**, **Lognormal**, **GEV**, **Gamma**, **Pearson type III** and **Log-Pearson type III**, the first requirement is that the parameters estimators must be **asymptotically efficient**, usually estimated using the **maximum likelihood method**. For the **Fréchet**, **Lognormal** and **Log-Pearson type III** the annual series must be transformed taking logarithms and then fitted the transformed series to the **Gumbel (max)**, **Normal** and **Pearson type III** distributions, respectively. After adjusting the data to any of these distributions using the **maximum likelihood method** the test statistics A^2 is estimated using:

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i - 1) [\ln(F_0(x_i)) + \ln(1 - F_0(x_{n-i+1}))]$$

where x_i are sorted in ascending order and $F_0(x_i)$ is the **cdf** evaluated in x_i . This is followed by the estimation of the parameter ω :

$$\omega = 0.116 \left(\frac{A^2 - \xi_n}{\beta_n} \right)^{1.1751\eta_n} + 0.0403 \quad \text{for } \xi_n \leq A^2$$
$$\omega = \left[0.116 \left(\frac{0.2\xi_n}{\beta_n} \right)^{1.1751\eta_n} + 0.0403 \right] \left[\frac{A^2 - 0.2\xi_n}{\beta_n} \right] \quad \text{for } \xi_n > A^2$$

where the parameters ξ_n , β_n and η_n are computed using the expressions in the following table for different distributions and according to the size n of the annual series.

Anderson-Darling test

Tabla 2.30: Parámetros para calcular ω

Distribución	ξ_n	β_n	η_n
Gumbel / Frechét	$0,169 \left(1 + \frac{0,1}{n}\right)$	$0,229 \left(1 - \frac{0,2}{n}\right)$	$1,141 \left(1 + \frac{0,5}{n}\right)$
Normal y Lognormal	$0,167 \left(1 + \frac{0,3}{n}\right)$	$0,229 \left(1 - \frac{0,2}{n}\right)$	$1,147 \left(1 + \frac{0,5}{n}\right)$
GEV	$0,147 (1 - 0,13b + 0,21b^2 + 0,09b^3) \times \left(1 + \frac{0,9}{n} - \frac{0,2}{\sqrt{n}}\right)$	$0,189 (1 + 0,2b + 0,37b^2 + 0,17b^3) \times \left(1 - \frac{1,8}{n}\right)$	$1,186 (1 - 0,04b - 0,04b^2 - 0,01b^3) \times \left(1 - \frac{0,7}{n} + \frac{0,2}{\sqrt{n}}\right)$
Gamma	$0,145 (1 + 0,17r^{-1} + 0,33r^{-2}) \times \left(1 + \frac{2,0}{n} - \frac{0,3}{\sqrt{n}} - \frac{0,4}{r\sqrt{n}}\right)$	$0,186 (1 + 0,34r^{-1} + 0,3r^{-2}) \times \left(1 - \frac{0,5}{n} - \frac{0,3}{\sqrt{n}} + \frac{0,3}{r\sqrt{n}}\right)$	$1,194 (1 - 0,04r^{-1} - 0,12r^{-2}) \times \left(1 - \frac{1,8}{n} + \frac{0,1}{\sqrt{n}} + \frac{0,5}{r\sqrt{n}}\right)$
Pearson Tipo III Log Pearson Tipo III	$0,145 (1 + 0,17b^{-1} + 0,33b^{-2}) \times \left(1 + \frac{2,0}{n} - \frac{0,3}{\sqrt{n}} - \frac{0,4}{b\sqrt{n}}\right)$	$0,186 (1 + 0,34b^{-1} + 0,3b^{-2}) \times \left(1 - \frac{0,5}{n} - \frac{0,3}{\sqrt{n}} + \frac{0,3}{b\sqrt{n}}\right)$	$1,194 (1 - 0,04r^{-1} - 0,12b^{-2}) \times \left(1 - \frac{1,8}{n} + \frac{0,1}{\sqrt{n}} + \frac{0,5}{b\sqrt{n}}\right)$

Para GEV: si $b > 0,5$, hacer $b = 0,5$

Para Gamma o Log Pearson Tipo III: si $r \circ b < 2$, hacer $r \circ b = 2$

Basado en Laio (2004)

Anderson-Darling test

The **Anderson-Darling test** shows that if $\omega < \omega_*$, where $\omega_* = 0.743, 0.581, 0.461$ and 0.347 for significant levels of $\alpha = 0.01, 0.025, 0.05$ and 0.1 , H_0 is accepted, which means that the annual series can be represented by the chosen distribution. Alternatively, with the value of ω , F is calculated as as:

$$F(\omega) = \frac{1}{\pi\sqrt{\omega}} \left[e^{(-\frac{1}{16\omega})} K_{1/4} \left(\frac{1}{16\omega} \right) \right] + \frac{1}{\pi\sqrt{\omega}} \left[1.118e^{(-\frac{25}{16\omega})} K_{1/4} \left(\frac{25}{16\omega} \right) \right] \text{ for } \omega < 1.2$$

$$F(\omega) = 1 \text{ for } \omega \geq 1.2$$

where $K_{1/4}[\cdot]$ is the modified **Bessel function** of order $1/4$. If $F(\omega) < (1 - \alpha)$, H_0 is accepted; the lesser $F(\omega)$ the more suitable is the distribution chosen.

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

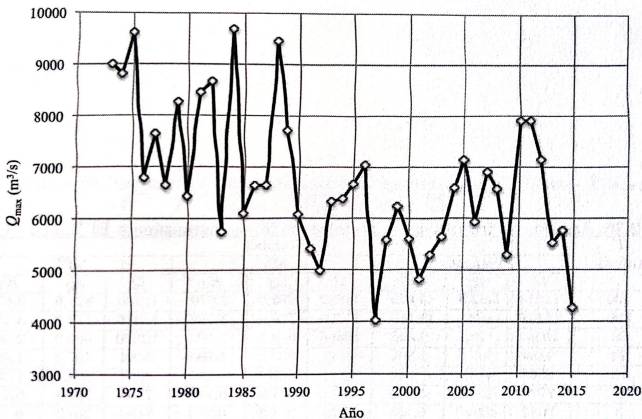


Figura 2.5: Serie de tiempo de la serie anual de caudales máximos instantáneos en El Banco

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

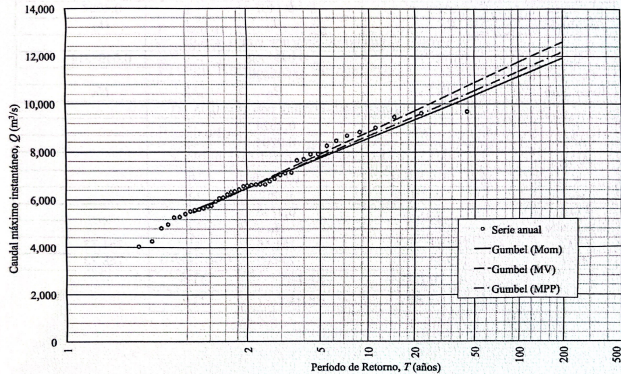


Figura 2.6: Curvas de frecuencia estimadas con Gumbel para la serie anual de caudales máximos instantáneos en El Banco

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

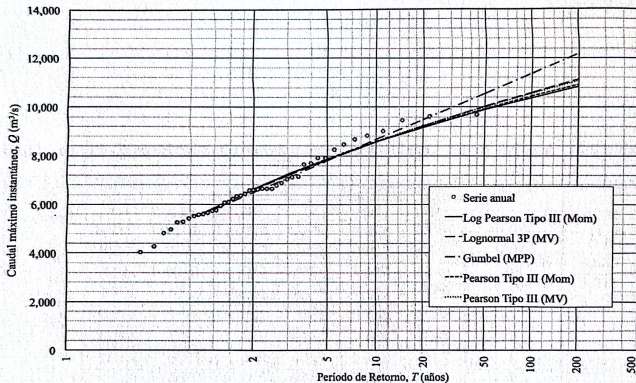


Figura 2.7: Curvas de frecuencia estimadas con Pearson y Log Pearson Tipo III y Lognormal 3P para la serie anual de caudales máximos instantáneos en El Banco

Distribution fitting

Annual series of maximum streamflow at Banco station, Magdalena River

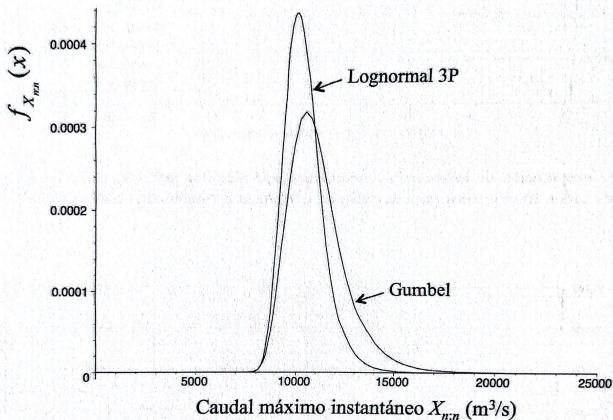


Figura 2.15: Funciones de densidad de probabilidad $f_{X_{n:n}}(x)$ del máximo caudal en 43 años en El Banco, a partir de las curvas de frecuencia ajustadas con Gumbel (MPP) y Lognormal 3P (MV)

Distribution selection

Generalities

- ▶ The distribution fitting processes described above can be applied to different types of distributions using different parameter-estimation methods.
- ▶ According to the literature, there is not one 'perfect' distribution for a specific study case, so multiple **pdfs** must be tried and initially analysed based on **plotting** and **hypothesis testing**.
- ▶ There are two types of procedures to select **probability distribution**: 1) **Goodness-of-fit tests** (e.g. Chi-squared and Kolmogorov-Smirnov goodness-of-fit tests), 2) **information theory procedures** (e.g. Akaike information criteria (AIC) and Bayesian information criteria (BIC)).

Distribution selection

AIC criteria

The **AIC** criteria is defined as

$$AIC = -2 \ln \left[\prod_{i=1}^n f_X(x_i, \hat{\theta}) \right] + 2p$$

where p is the number of parameters of $f_X(x)$, $\hat{\theta}$ represents the fitted values of the parameters, n is the size of the annual series and x_i is each value of the series. The first term of the right hand side of the equation represent the likelihood function and it is a measure of the inaccuracy of the estimation, while the second term account for the uncertainty in the estimation due to the number of parameters considering the distribution parsimony effects. The lesser AIC the better $f_X(x)$. The equation above for AIC is better for annual time series with $n \geq 30$. When $n/p < 40$, a corrected AIC must be calculated as:

$$AIC = -2 \ln \left[\prod_{i=1}^n f_X(x_i, \hat{\theta}) \right] + 2p \left(\frac{n}{n - p - 1} \right)$$

BIC criteria

The **BIC** criteria result from a Bayesian analysis, so it is different to the **AIC criteria**. The **BIC criteria** is defined as

$$BIC = -2 \ln \left[\prod_{i=1}^n f_X(x_i, \hat{\theta}) \right] + \ln(n)p$$

The lesser BIC the better $f_X(x)$. The **BIC criteria** is more adequate for parsimonious distributions than the **AIC criteria** when $n \geq 8$.

Tail classes

Regarding that one of the objective of the frequency analysis of extreme events is to estimate **quantiles** associated to high return periods, the fitted distributions must guarantee that their tails are well represented (inferior tail for minimum values and superior tails for maximum values) because extreme events are associated to tails. According to the tail shape, a distribution can have **light tails** if the probability of extreme values decreases very quickly, and **heavy tails** if the probability of extreme values (far from the mean) is relatively high. This means that for a **heavy tail** distribution, the magnitude is more extreme for the same return period. So before fitting different distributions is necessary to make exploratory analysis to identify the type of tail.

Distribution selection

Tail classes

The following methodologies serve to identify tails:

1. Analyse the **histogram** of the data sample, once the sample is cleared up of outliers, the extreme values and the **Q-Q plot**; this last plots magnitudes from the observed and the theoretical distribution for different quantiles. The **Q-Q plot** is made for the observed magnitude vs each theoretical distribution. Note that if the **Q-Q** is lineal, the theoretical distribution is correct. In contrast, if the **Q-Q plot** is not lineal, this means that the annual time series have lighter or heavier tail than the theoretical distribution.

2. The **mean excess function**
$$e_i = \frac{\sum_{j=1}^n (x_j - x_i) I_{(x_j > x_i)}}{\sum_{j=1}^n I_{(x_j > x_i)}}$$

where $I_{(x_j > x_i)} = 1$ if $x_j > x_i$ and 0 otherwise, with the data sorted in descending order. So, e_i describe the expected excess over x_i . The plot i vs e_i is the empirical **mean excess function**. If the **slope** in this plot is horizontal, the data sample follows a **exponential distribution**, if the **slope** is positive the data follow a **heavy tail distribution** and if the **slope** is negative the data is associated to a **light tail distribution**.

3. The **Hill's plot** is based on the following estimator

$$\xi_i = \frac{1}{i-1} \left[\sum_{j=1}^{i-1} \ln x_j \right] - \ln x_i$$

with the data sorted in descending order and valid for $i \geq 2$. The **Hill's plot** is i vs ξ_i . If the plot tends to a constant value different of 0 the distribution tail is heavy, if the plot tends to 0 the distribution tail tends to be light as in the **exponential distribution**.

Distribution selection

Classification of asymptotic tail behaviour for some pdfs

Clase	Tipo de cola	Características	Fdps	Comp. asintót.
A	Muy pesada	Cola de Pareto $0 < \alpha < 2$ Varianza infinita		$x \propto T^p$
B	Pareto	$F_X(x) = 1 - (u/x)^\alpha$ $\alpha > 0$ Cola decae con ley de potencia k momentos acotados ($k < \alpha$)	Pareto I Pareto II GPD	$x \propto T^p$
C	Regularmente variable	Cola no es de Pareto pero es similar a Pareto, y cae asintóticamente con ley de potencia	Fréchet Halphen inv. Tipo B Gamma inv. Log Pearson Tipo III Gamma Generalizada	$x \propto T^p$
Subclase Log-normal		Asintóticamente declina de forma similar a clases C o C/D	Lognormal 2P Lognormal 3P	$x \propto e^{(\ln T)^{0.5}}$
D	Subexponencial	Cola decrece más lentamente que la Clase E	Gumbel Halphen Tipo A Pearson Tipo III Gamma Halphen Tipo B	$x \propto (\ln T)^p$
E	Exp.	fdps con $E[e^{sX}] = \infty$	Exponencial	$x \propto (\ln T)^p$
Normal		Cola más liviana que la Clase E	Normal	

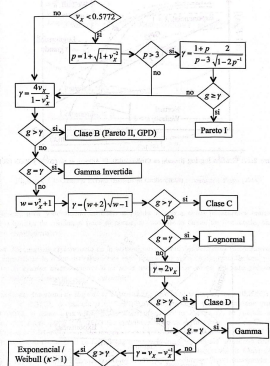
Basado en El Adlouni et al. (2008) y Papalexiou et al. (2013)

Distribution selection

Procedure to select the distribution

The following is a procedure to select the most adequate distributions for the annual series

1. The flow diagram below is based in the theoretical behaviour of the first three **moments** of the distributions, where v_X and g are the coefficients of **variation** and **asymmetry** of the annual series, respectively. Based on these values is possible to identify the distribution that fit the best to the annual series.



2. Some discriminatory plots based on the annual series:

- **log-log plot**: Plot of the $\log x_i$ vs $\log p_i$, where p_i the exceedance probability. If the plot is lineal, the distribution is C class; if the plot is concave is D or E class.
- **lognormality plot**: if $10v_X - 1.4 > g$ is suggested to plot the **cdf** of the annual serie with the normal probability in the horizontal axis and $\log x_i$ in the vertical axis. If the plot is lineal, the best distribution is **lognormal** and **hypothesis testing** of lognormality are recommended.
- **mean excess function**: If the **slope** tends to zero, the distribution is E class, if the **slope** is positive is D class. In this plot is recommended to plot those values greater than the median.

Distribution selection

Procedure to select the distribution

- ▶ **Hill's plot:** If the curve decrease asymptotically toward 0, the distribution is D or E class. However, if the plot converges to a different value, the distribution is C class.
- ▶ **Zenga plot:** This plot is useful to distinguish between **lognormal** and **pareto** distributions, where the n values of the annual series are sorted in ascending order. The x_i values are also grouped according to s values, so for instance, for x_j , $x_1 < \dots < x_j < \dots < x_s$ with frequencies n_j (number of times x_j is repeated in the annual series). So for $j = 1, \dots, s$, the following variables are defined:

$$N_j = \sum_{i=1}^j n_j \quad u_j = \frac{N_j}{n} \quad T_j = \sum_{i=1}^j x_i n_i \quad T = \sum_{j=1}^s x_j n_j$$
$$Q_j^- = \frac{T_j}{N_j} \quad Q_j^+ = \begin{cases} \frac{T - T_j}{n - N_j} & j = 1, 2, \dots, s-1 \\ x_s & j = s \end{cases} \quad z_j = 1 - \frac{Q_j^-}{Q_j^+}$$

The **Zenga plot** is u_j vs z_j . If the **slope** is positive, the distribution could be B or C class, if the **slope** tends to be horizontal it can be lognormal class, and if it negative the distribution is D or E class.

Note that for frequency analysis of **minimum values** in the **lognormality plot** must be analysed the inferior tail (lefthand side tail), while for the **log-log**, the **Hill**, the **mean excess function** and the **Zenga** plots must be made making a reflection transformation of the data. The **reflection transformation** is made by doing $x_i = -x_i$.

Distribution selection

Procedure to select the distribution

After the transformation, for the frequency analysis of **minimum values** in the **log-log plot** the horizontal axis correspond to $-\ln(-x_i)$ and the vertical axis correspond to

$\ln[1 - F_{-X}(-x_i)]$. An for the **Zenga plot** $z_j = \frac{Q_j^-}{Q_j^+}$.

Once the most appropriate **pdfs** are identified, they are fitted to the data sample using the methods describe before and assessing the fitting using the **AIC** and **BIC** criteria. If there are more than one **pdf**, using the **BIC** value for each distribution, it is possible to estimate a weighted x_T value for different return periods (T) from the **frequency curve**. Based on the **BIC** values, posterior probabilities (b_j) are estimated to compute the weights (w_j). The weighted x_T are calculated as

$$w_j = \frac{b_j}{\sum_{i=1}^m b_i} \quad b_j = e^{-0.5BIC_j} \quad x_T = \sum_{j=1}^m w_j x_{T_j}$$

where m is the number of selected **pdfs**.

This procedure can be applied also to parameters such the s_X or the **confidence limits** to obtain a weighted version of them.